

A MODIFIED GMRES METHOD FOR SOLVING LARGE NONSYMMETRIC LINEAR SYSTEMS*

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Abstract A modified GMRES method is proposed in this paper, the method replaces the approximation x_m obtained by the GMRES method with a new approximation $\tilde{x}_m$ which is a linear combination of x_m and the wasted basis vector v_{m+1} . The residual norm of the new approximation satisfies a small one-dimensional minimization problem. Relationships between the residual norms of x_m and $\tilde{x}_m$ are given. We show that the resulting m -step modified GMRES method is better than the original m -step GMRES method in theory and is considerably more efficient than the $(m+1)$ -step GMRES method by numerical experiments.

Key words linear systems, Krylov subspace, Arnoldi process, the GMRES method.

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1 Introduction

The conjugate gradient method (CG) is one of the most popular iterative methods for solving symmetric positive definite linear systems. For indefinite symmetric linear systems, Paige and Saunders showed that CG method can be generalized to find the minimum residual solution at every step of the iteration. The resulting method is called the minimum residual method (MINRES) [7]. To dealing with large nonsymmetric linear systems, several different generalizations of CG have been proposed, see, for example [2,3,8,10]. Among them, one of the best known method is generalized minimum residual method (GMRES) proposed by Saad and Schultz [8]. The GMRES method is a Krylov subspace based method. It relies on the Arnoldi process, and the approximate solution has the property of minimizing at every step the norm of the residual vector over the Krylov subspace. Therefore it can viewed as direct generalization of the MINRES method to deal with nonsymmetric linear systems.

A variety of interesting implementations of the GMRES method have been developed in [11,12]. They mainly concern on using reliable technique to construct an orthornormal basis of the Krylov subspace. In this paper, a different approach is proposed by our observation that the $(m+1)$ -th Arnoldi vector v_{m+1} is already available when the m steps of the Arnoldi process are performed. However, the approximate solution x_m formed is in $\mathcal{K}_m(A, v_1)$ rather than in $\mathcal{K}_{m+1}(A, v_1)$. That is, though v_{m+1} has already generated, it contributes nothing

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to the approximate solution in the original GMRES method. In order to make use of the available v_{m+1} , we propose a new strategy that replaces x_m with $\tilde{x}_m$, which is a linear combination of x_m and v_{m+1} . And $\tilde{x}_m$ satisfies a one-dimensional minimization problem. The resulting method is named the modified GMRES method. The total extra costs of the new method is one matrix-vector multiplication plus two inner products. So the resulting m -step modified GMRES method is cheaper than the original $(m+1)$ -step one, which requires extra one matrix-vector multiplication and at least m inner products. Similar strategy was originally proposed by Jia in solving large nonsymmetric eigenproblems [4].

In section 2, we give a brief review of the GMRES method. We present the modified GMRES method in section 3. Then numerical results are reported in section 4. Comparisons between the standard $(m+1)$ -step GMRES and the m -step modified GMRES show that the later one is often considerably more efficient.

Throughout the paper, denote by $\mathcal{K}_m(A, v_1) = \text{span}\{v_1, Av_1, \dots, A^{m-1}v_1\}$ the m -dimensional Krylov subspace generated by A and v_1 , f^* denotes the conjugate transpose of vector f , by $\mathcal{C}^m$ the m -dimensional complex space and by e_m the m th coordinate vector of dimension m . The norm $\|\cdot\|$ denotes both the Euclidean vector norm and the subordinate spectral matrix norm.

2 The GMRES Method

Consider large linear system

$$Ax = b, \quad (1)$$

where A is an $n \times n$ nonsingular matrix, and b is vector of length n . The GMRES method is an iterative method in nature, it generates a sequence of approximations converging to the solution $\hat{x}$ of (1). The approximations have the property of minimizing at every step the norm of the residual vector over a Krylov subspace.

Suppose x_0 is an initial guess to the solution $\hat{x}$, and $r_0 = b - Ax_0$ is the initial residual vector. The GMRES method seeks an approximate solution of the form

$$x_m = x_0 + V_m y_m, \quad (2)$$

where columns of V_m form an orthonormal basis for the Krylov subspace of dimension m defined by $\mathcal{K}_m(A, r_0)$

$$\mathcal{K}_m(A, r_0) = \text{span}\{r_0, Ar_0, \dots, A^{m-1}r_0\}. \quad (3)$$

The vector y_m is determined so that the norm of the residual $r_m = b - Ax_m$ is minimal over $\mathcal{K}_m(A, r_0)$. The orthonormal basis $v_1, \dots, v_m$ for the Krylov subspace $\mathcal{K}_m(A, r_0)$ is obtained via the well known Arnoldi process, which satisfies the relationship

$$AV_m = V_m H_m + h_{m+1,m} v_{m+1} e_m^* = V_{m+1} \tilde{H}_m, \quad (4)$$

where $\tilde{H}_m$ is the $(m+1) \times m$ upper Hessenberg matrix that is the same with H_m except for an additional row whose only nonzero entry is $h_{m+1,m}$ in the position $(m+1, m)$.

Let $v_1 = r_0/\beta$ where $\beta = \|r_0\|$. The residual r_m associated with the approximate

solution x_m satisfies

$$\begin{aligned} r_m &= b - Ax_m = r_0 - AV_m y_m \\ &= r_0 - V_{m+1} \tilde{H}_m y_m \\ &= \beta v_1 - V_{m+1} \tilde{H}_m y_m \\ &= V_{m+1} (\beta e_1 - \tilde{H}_m y_m). \end{aligned}$$

Since V_{m+1} is a matrix with orthonormal columns, so the residual norm $\|r_m\| = \|\beta e_1 - \tilde{H}_m y_m\|$ is minimal when y_m solves the following least square problem

$$\min_{y \in \mathbb{C}^m} \|\beta e_1 - \tilde{H}_m y\|. \quad (5)$$

The following is a version of the restarted GMRES algorithm.

Algorithm 1 The restarted m-step GMRES algorithm

1. Start: Choose a stopping criterion ϵ and an initial guess x_0 , compute $r_0 = b - Ax_0$ and $v_1 = r_0 / \|r_0\|$. Set the size of the Krylov subspace to be m .

2. Iteration:

for $j = 1 : m$ do: $v = Av_j$;

for $i = 1 : j$ do:

$h_{i,j} = (v, v_i)$;

$v = v - h_{i,j} v_i$;

end

$h_{j+1,j} = \|v\|$;

$v_{j+1} = v / h_{j+1,j}$;

end

3. Form the approximate solution: Compute y_m from (5), form $x_m = x_0 + V_m y_m$.

4. Restart: Compute the residual vector $r_m = b - Ax_m$, and relative residual norm $\frac{\|r_m\|}{\|b\|}$, if satisfied then stop, else choose a new x_0 and go to 1.

The details on the practical implementation of the algorithm above are discussed at length in [8].

3 A Modified GMRES Method

Rather than using $x_m = x_0 + V_m y_m$ as an approximate solution, we now choose $\tilde{x}_m$ of the form

$$\tilde{x}_m = x_m + \alpha_m v_{m+1} \quad (6)$$

as a modified approximation to $\hat{x}$. Where α_m is the solution of the following minimization problem

$$\min_{\alpha \in \mathbb{C}} \|b - A(x_m + \alpha v_{m+1})\|. \quad (7)$$

Theorem 1 Define $r_m = b - Ax_m$, $f = Av_{m+1}$ and $\tilde{r}_m = b - A\tilde{x}_m$, where $\tilde{x}_m$ is given

by (6), α_m solves $\min_{\alpha \in C} \|b - A(x_m + \alpha v_{m+1})\|$. Then

$$\alpha_m = \frac{f^* r_m}{\|f\|^2}. \quad (8)$$

$$\|\tilde{r}_m\| = \frac{\sqrt{\|r_m\|^2 \|f\|^2 - |f^* r_m|^2}}{\|f\|}. \quad (9)$$

Proof As

$$\begin{aligned} b - A\tilde{x}_m &= b - Ax_m - \alpha Av_{m+1} \\ &= r_m - \alpha f, \end{aligned}$$

Therefore, we obtain from (7) that

$$\min_{\alpha \in C} \|b - A(x_m + \alpha v_{m+1})\| = \min_{\alpha \in C} \|r_m - \alpha f\|.$$

It is easy to see that solution of the above minimization problem is

$$\alpha_m = \frac{f^* r_m}{\|f\|^2}.$$

Hence, we have

$$\begin{aligned} \|\tilde{r}_m\| &= \min_{\alpha \in C} \|b - A(x_m + \alpha v_{m+1})\| \\ &= \|(I - (\frac{f}{\|f\|})(\frac{f}{\|f\|})^*)r_m\| \\ &= \frac{\sqrt{\|r_m\|^2 \|f\|^2 - |f^* r_m|^2}}{\|f\|}. \end{aligned}$$

Remark

(1) From (9) it is easy to see that $\|\tilde{r}_m\| < \|r_m\|$ whenever $f^* r_m \neq 0$. So $\tilde{x}_m$ is at least as good as x_m .

(2) Note that $\|\tilde{r}_m\| = \|(I - (\frac{f}{\|f\|})(\frac{f}{\|f\|})^*)r_m\|$ is the distance between r_m and the subspace spanned by $\{f\}$. therefore, the smaller the distance is, the smaller $\|\tilde{r}_m\|$ is.

Based on the above analysis, we present the following algorithm.

Algorithm 2 A modified m-step GMRES algorithm

1. Start: Choose a stopping criterion ϵ and an initial guess x_0 , compute $r_0 = b - Ax_0$ and $v_1 = r_0/\|r_0\|$. Set the size of the Krylov subspace to be m .

2. Iteration:

for $j = 1 : m$ do:

$v = Av_j$;

for $i = 1 : j$ do:

$h_{i,j} = (v, v_i)$;

$v = v - h_{i,j}v_i$;

end

$$h_{j+1,j} = \|v\|;$$

$$v_{j+1} = v/h_{j+1,j};$$

end

3. Form the approximate solution: Compute $\tilde{x}_m$ by (5)(6), and (8).

4. Restart: Compute the residual vector $\tilde{r}_m = b - A\tilde{x}_m$ and the relative residual norm $\frac{\|\tilde{r}_m\|}{\|b\|}$, if satisfied then stop, else choose a new x_0 and go to 1.

Remark

1. In practice, Iterative refinement is needed in step 2 of Algorithm 1 and Algorithm 2 in order to compensate for any loss in orthogonality.

2. In practical implementations, the option of choosing the new starting vector x_0 in step 4 of the above two algorithms are various. This paper adopts the strategy proposed by Saad and Schultz [8], i.e. the new starting vector to be the normalized residual vector.

4 Numerical Experiments

In this section the modified GMRES method (MGMRES) is tested and compared with GMRES method. All numerical tests have been carried out on an INTEL PENTIUM IV 2.4GHz with main memory 256MB and the machine precision $eps = 2.22 \times 10^{-16}$ using MATLAB 6.5 on a Window XP-based system. In all tables of the paper, m is the dimension of the Krylov subspace, $iter$ denotes the number of restarts, cpu is the CPU timings in seconds, mv is the matrix-vector products accessing the matrix A , res is the relative residual norm. In the following examples we set the stopping criterion to be $1E-10$. GMRES(m) denotes the m -step GMRES method, and MGMRES(m) denotes the m -step modified GMRES method.

Example 1 The first test problems are from Morgan [?]. The matrices are bidiagonal with 0.1 in each superdiagonal position. In Problem 1, the matrix has entries $1, 2, \dots, 1000$ in the main diagonal. For Problem 2, the diagonal entries are $-2, -1, 1, 2, \dots, 997, 998$.

Table 1 and table 2 show the results for the two test problems. For Problem 1, we can see that GMRES(10) needs 770 matrix-vector products to reach a relative residual norm of $9.9E-11$ whereas the MGMRES(9) needs only 671 matrix-vector products to reach a residual norm of $9.6E-11$. The matrix in Problem 3 is indefinite and the results show that after 251 restarts the relative residual norm of MGMRES(19) reaches $9.8E-11$, whereas GMRES(20) needs 273 restarts to reach the same relative residual norm. We remark that in both cases the CPU timings required by MGMRES(m) is less than GMRES(m) to reach the same residual norm.

Example 2 The second tests problems are taken from [5], The matrix A and the solution vector x are of the form

$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & \alpha \\ 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

The right hand side vector is set to $b = Ax$. For first example TP1, the parameters n, α are set to $n = 1000, \alpha = 20000$, the second example TP2 uses $n = 1000, \alpha = 1.1$.

In this example, TP1 denotes a high nonsymmetric matrix, and TP2 is nearly a sym-

metric one. we can see from Table 3 that in both cases, the amount of work and CPU timings required by MGMRES(9) are less than GMRES(10).

Example 3 This example is from the Harwell-Boeing sparse matrix collection [1]. Both sherman 1 and saylr 3 are nonsymmetric matrix of dimension 1000 and has 3750 nonzeros entries. Sherman 4 is also nonsymmetric, has dimension 1104 and 3876 nonzeros entries.

Table 1 Problem 1

	iter	cpu	mv	res
GMRES(10)	70	10.2	770	9.9E-11
MGMRES(9)	61	8.4	671	9.6E-11

Table 2 Problem 2

	iter	cpu	mv	res
GMRES(20)	273	65.6	5733	9.8E-11
MGMRES(19)	251	58.6	5271	9.8E-11

Table 3 TP(1000,20000) and TP(1000,1.1)

problem	method	iter	cpu	mv	res
TP1	GMRES(10)	66	9.6	726	9.8E-11
	MGMRES(9)	57	7.9	627	9.9E-11
TP2	GMRES(10)	68	9.9	748	8.9E-11
	MGMRES(9)	63	8.6	693	9.8E-11

Table 4 sherman1, sherman4 and saylr3

problem	method	iter	cpu	mv	res
sherman1	GMRES(10)	943	138.8	10373	9.8E-11
	MGMRES(9)	810	112.8	8910	9.4E-11
sherman4	GMRES(10)	120	20.4	1320	8.8E-11
	MGMRES(9)	100	16.1	1100	9.8E-11
saylr3	GMRES(20)	235	57.1	4935	8.7E-11
	MGMRES(19)	221	52.4	4641	9.8E-11

The convergence behavior of the three test problems are reported in Table 4. As in the above two examples, The amount of work, the CPU timings and the number of iterations are compared, we observe that m -step MGMRES has a considerable improvement over $(m+1)$ -step GMRES one in most cases.

All the above numerical experiments have shown that that MGMRES(m) is superior to GMRES(m) for different nonsymmetric sparse matrices. Other experiments have also demonstrated the efficiency of the new method for relatively bigger m .

Remark This paper aims at showing superiority of MGMRES over GMRES, so we don't consider much about restarting and preconditioning aspects. The restarting strategy we adopt was originally proposed by Saad and Schultz [8]. Improved restarting strategy has been proposed by Morgan[6]. His work consists of saving the approximate eigenvectors of the matrix A corresponding to the smallest eigenvalues in magnitude and adding them to the new Krylov subspace. Moreover, preconditioning techniques for GMRES are also presented by several authors. Especially, the variant FGMRES of Saad [9] which allow changes in the preconditioning at every step. It would be interesting to apply these advanced accelerating

technique to the MGMRES method.

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